SCITECH GROUP · USC ISI

AI Workflows on Laguna using Pegasus

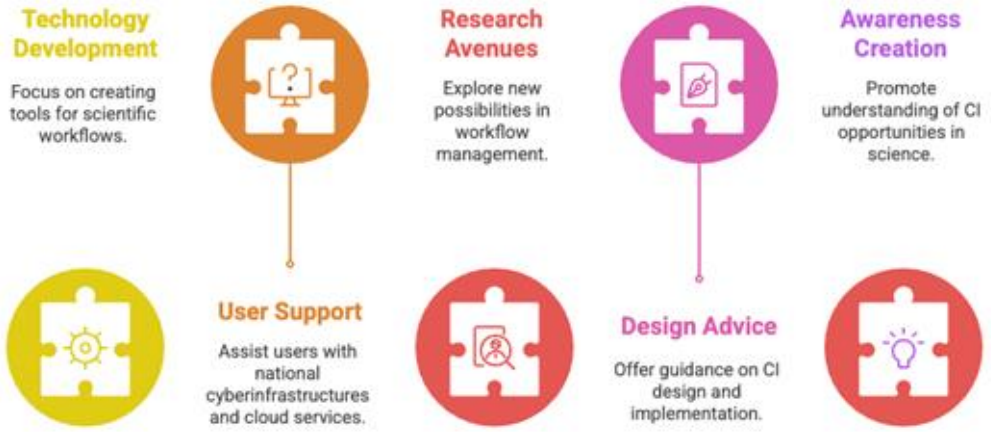
Karan Vahi · USC Information Sciences Institute

SciTech Activities

1. Develop technologies to support scientific workflows
2. Provide user support on national cyberinfrastructures and clouds
3. Research new avenues in the workflow management space
4. Provide advice on the design, implementation and use of Cyberinfrastructure (CI) for the Research Infrastructure lifecycle
5. Create awareness and understanding of opportunities in CI for Science



Marina del Rey, CA



Most projects are collaborative across 4-6 organizations

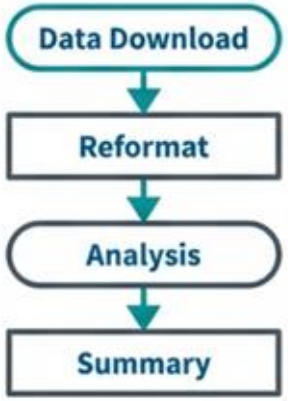
The Challenge: Moving Beyond the Laptop

The Scripting Trap



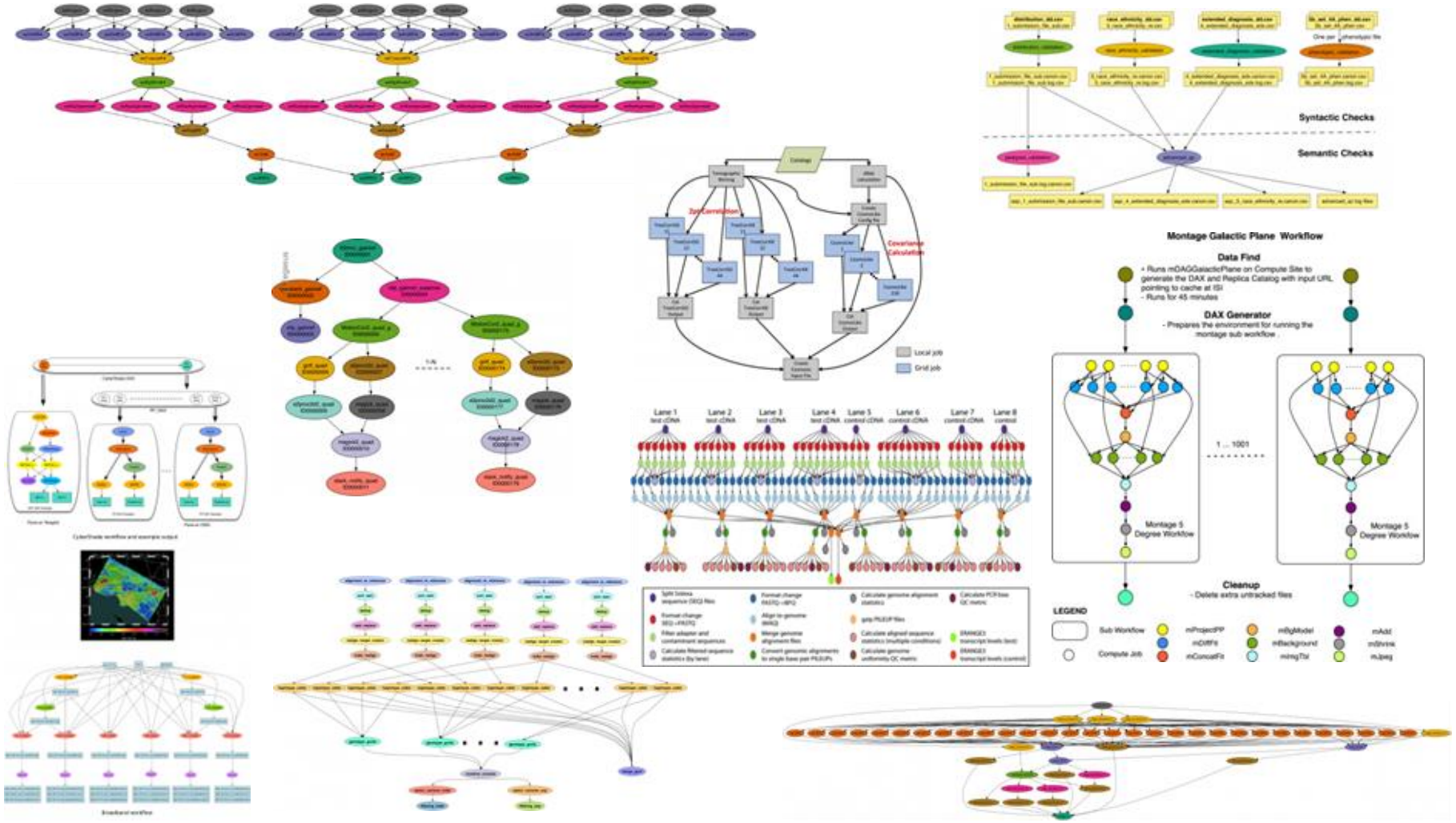
- Quick to code, but hard to scale
- Difficult to maintain and error-prone
- Hard to move between resources (Laptop vs. Cluster)
- “Works on my machine” syndrome

The Workflow Advantage



- Declarative: Describe what to do, not how
- Automated: Minimizes human involvement
- Provenance: Tracks history for reproducibility
- Portable: Runs on ACCESS, NAIRR, cloud, …

Workflows Built with Pegasus Across Science Domains



Pegasus Workflow Management System, est. 2001

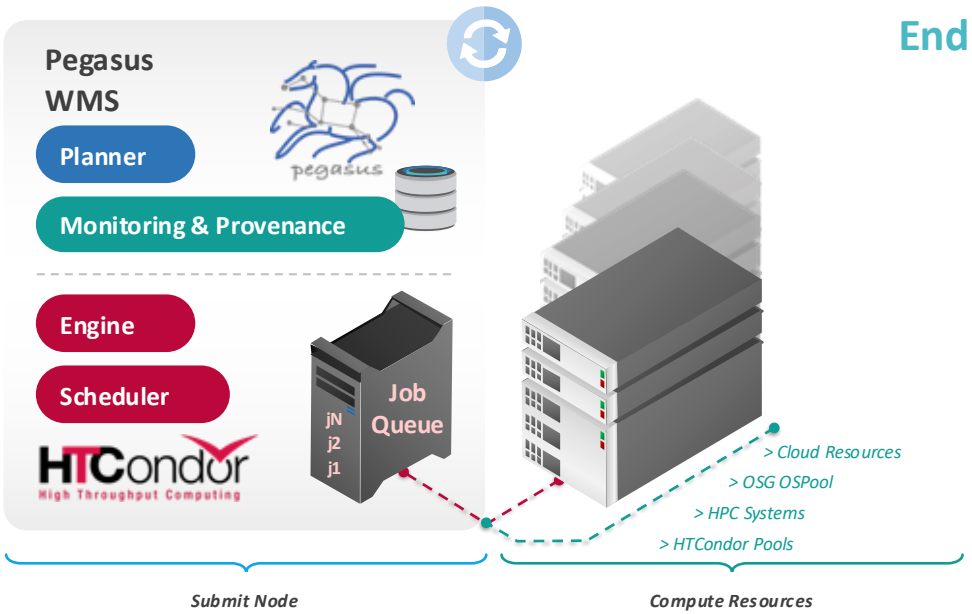
Workflow Challenges Across Domains

- Describe complex workflows in a simple way
- Access distributed, heterogeneous data and resources (heterogeneous interfaces)
- Deal with resources/software that change over time
- Ease of use. Ability to monitor and debug large workflows

Our Focus

- Separation between workflow description and workflow execution
- Workflow planning and scheduling (scalability, performance)
- Task execution (monitoring, fault tolerance, debugging, web dashboard)
- Workflow optimization, restructuring for performance and fault tolerance.

Pegasus Workflow Management System



End to End Workflow Management & Execution

- Develop portable scientific workflows in Python and Java
- Compile workflows to be run on heterogeneous resources
- Monitor and debug workflow execution via CLI and web-based tools
- Recover from failures with built-in fault tolerance mechanisms
- Regular release schedule incorporating latest research and development

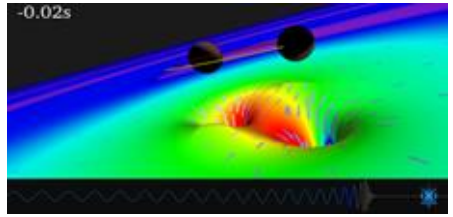
Pegasus in practice

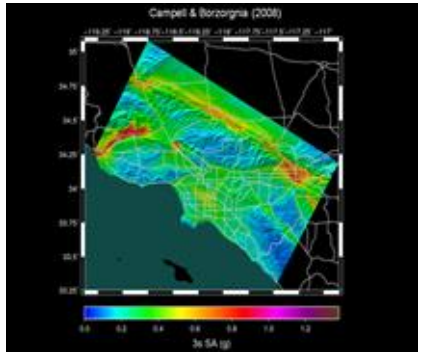
- Laser Interferometer Gravitational Wave Observatory (LIGO) develops large scale analysis pipelines used for gravitational wave detection.
- Southern California Earthquake Center (SCEC) CyberShake project generates hazard maps using hierarchical workflows .
- The Event Horizon Telescope uses Pegasus to orchestrate and scale complex distributed computing pipelines for data calibration, massive synthetic data generation, and deep learning inference

Event Horizon Telescope



LIGO observation of colliding black holes





Hazard map indicating maximum amount of shaking at a particular geographic location generated from SCEC's CyberShake Pegasus workflow

Application Showcase: https://pegasus.isi.edu/application-showcase/

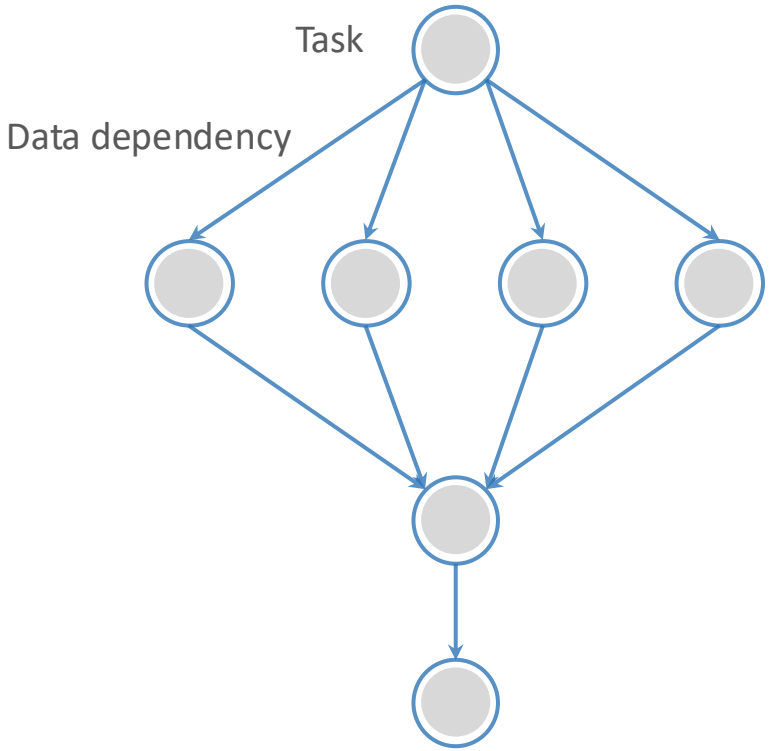
Key Pegasus Concepts

- **Pegasus WMS ==** **Pegasus planner (mapper) + DAGMan workflow engine + HTCondor scheduler/broker**
  - Pegasus maps workflows to infrastructure
  - DAGMan manages dependencies and reliability
  - HTCondor is used as a broker to interface with different schedulers

- **Workflows are DAGs**
  - Nodes: jobs, edges: dependencies
  - No while loops, no conditional branches
  - Jobs are standalone executables

- **Planning occurs ahead of execution**

- **Planning converts an abstract workflow into a concrete, executable workflow**
  - Planner is like a compiler



Defining Workflows in Python

```
from Pegasus.api import *

# Create abstract workflow
wf = Workflow("pipeline")

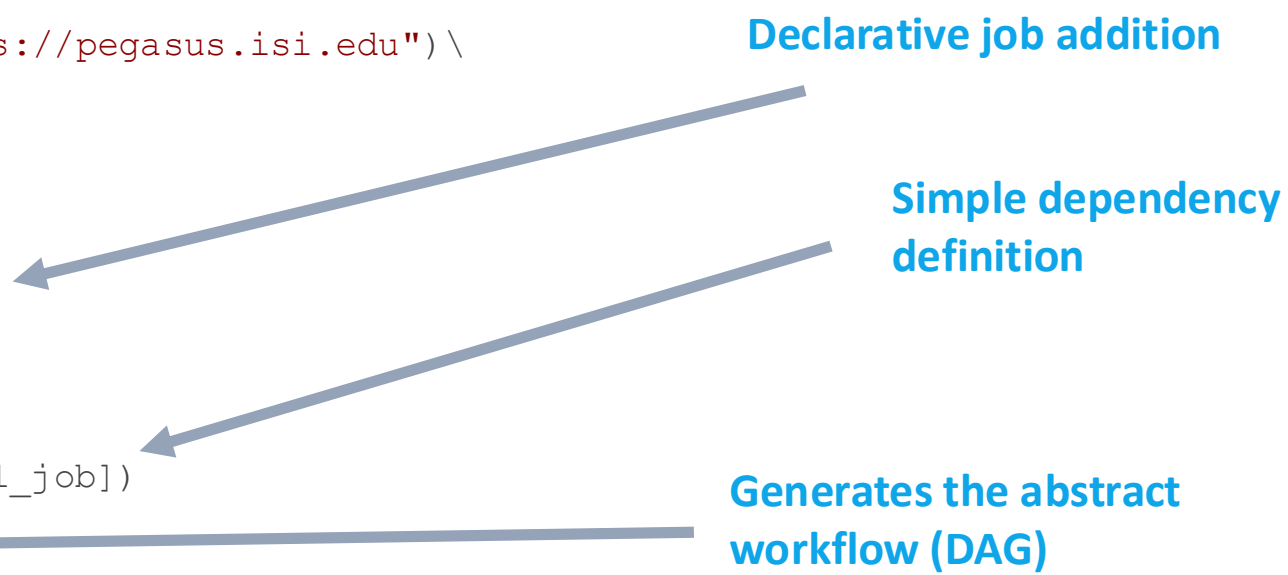
# Track the data
page = File("page.html")

# Create a Job
curl_job = Job("curl")\
           .add_args("-o", page, "https://pegasus.isi.edu")\
           .add_outputs(page)

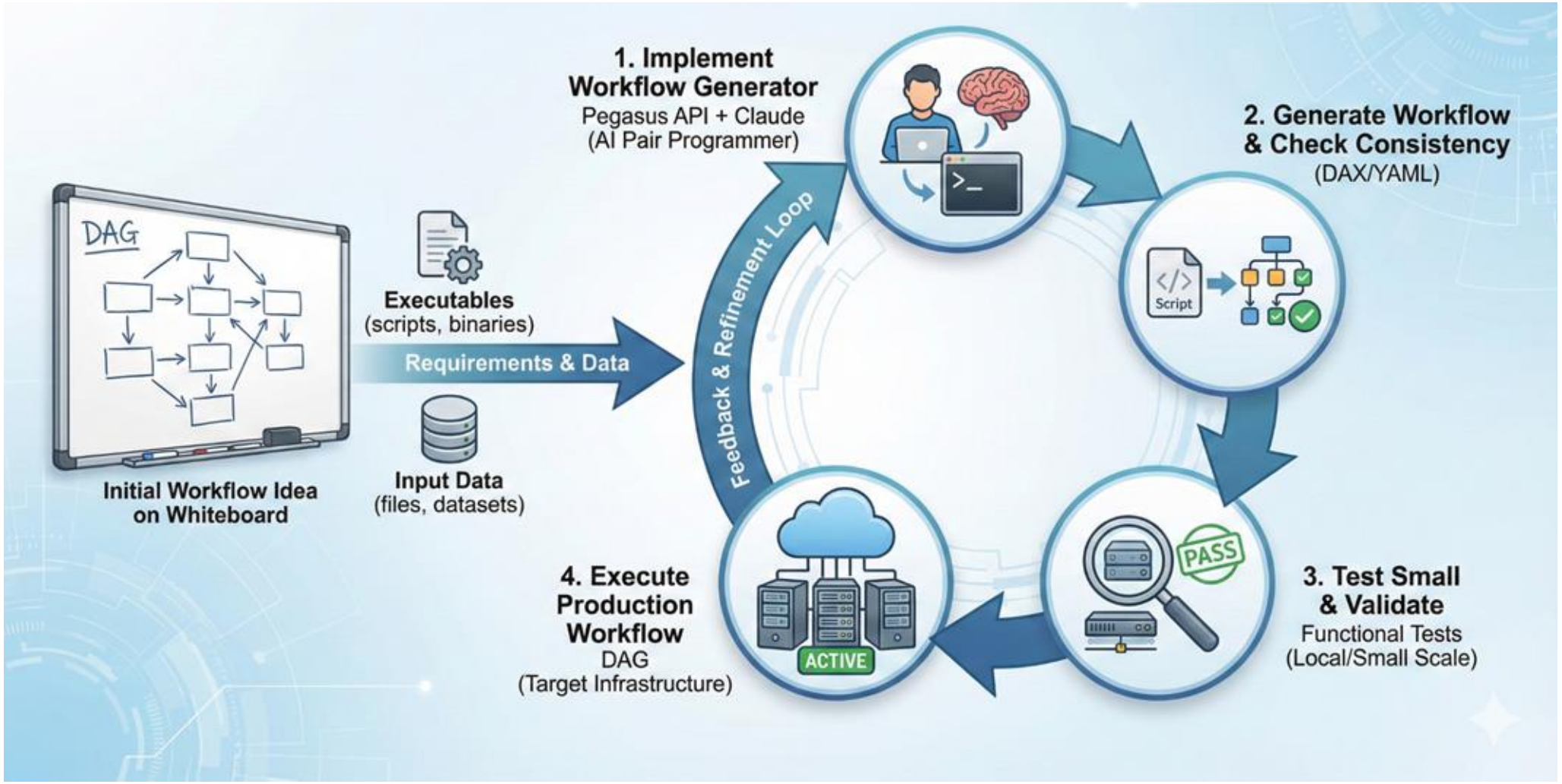
# Create a Dependent Job
wc_job = Job("wc")\
         .add_args("-l", page)\
         .add_inputs(page)

# Add jobs and dependencies
wf.add_jobs(curl_job, wc_job)
wf.add_dependency(wc_job, parents=[curl_job])

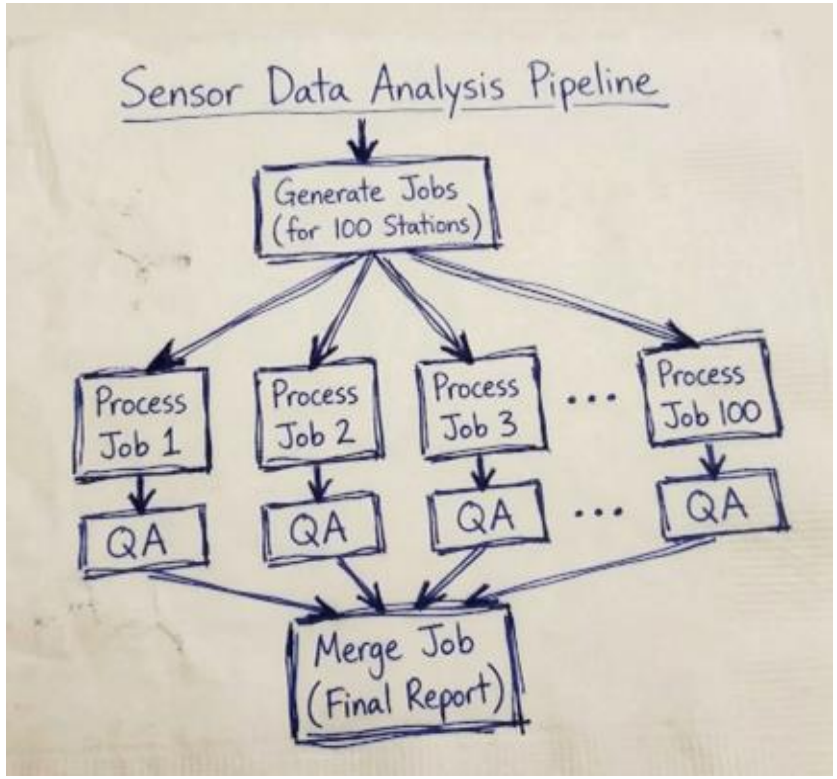
wf.write("pipeline.yml")
```



Designing Pegasus WMS Workflows with Claude



From Napkin Sketch to Executable Workflow



Claude reads the image directly

1. **Vision analysis**
   Identifies 4 processing stages, decision diamonds, fork/join points, data-flow arrows

2. **Topology extraction**
   Maps fan-out/fan-in DAG: generate → 100 parallel process+QA branches → merge report

3. **Code generation**
   Produces workflow_generator.py with Pegasus API calls, infer_dependencies=True

4. **Script scaffolding**
   Creates 4 standalone bin/ scripts — generate, process, QA, merge for 100 stations

5. **Container + tests**
   Generates Docker config, test_workflow.py, and parameterized station count

No manual transcription needed. The AI reads the visual spec the way a human collaborator would.

Domain scientists draw what they know — AI translates it into executable workflow code

More on this in the poster session on Pegasus AI Studio

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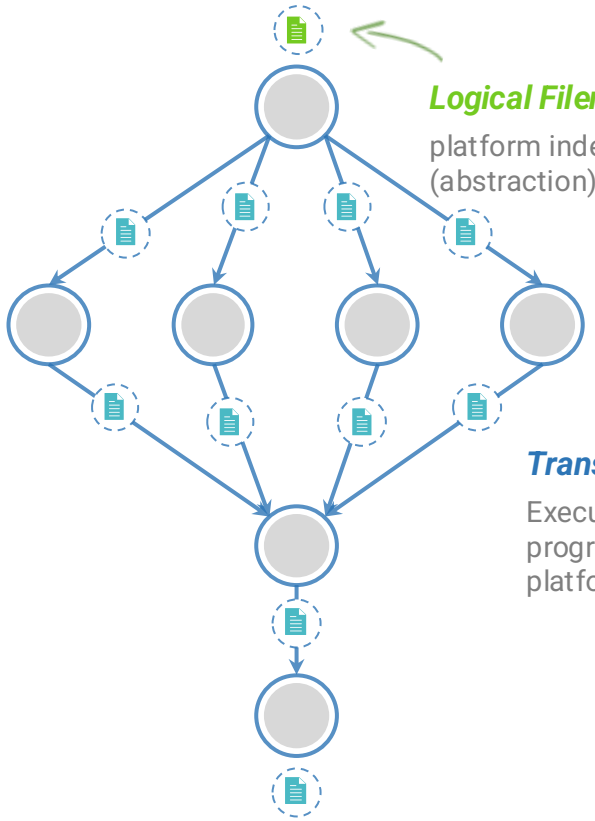
INPUT WORKFLOW SPECIFICATION

Portable Description

Users do not worry about low level execution details

YAML formatted

ABSTRACT WORKFLOW



Logical Filename (LFN)

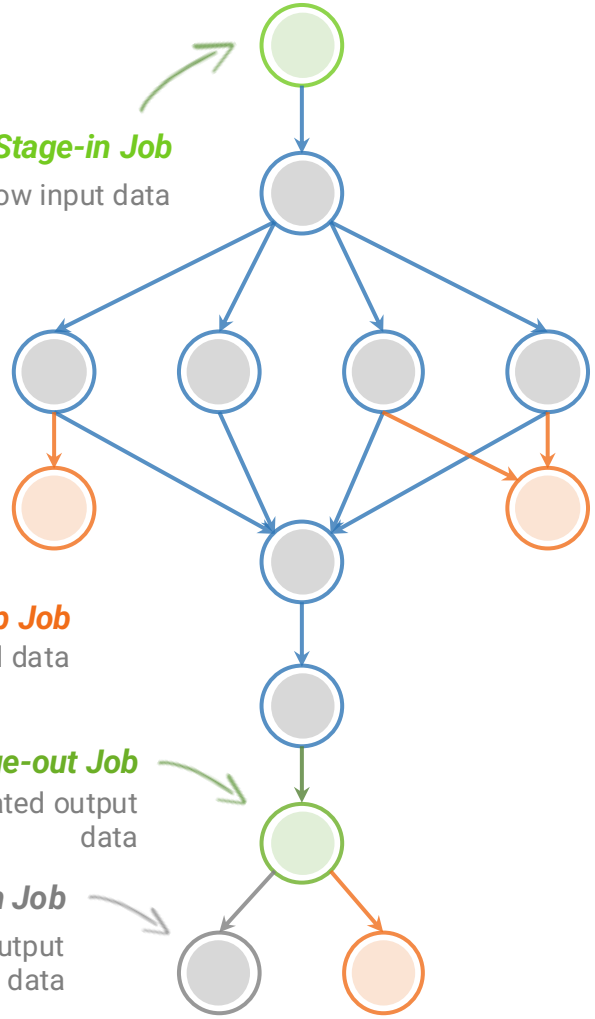
platform independent (abstraction)

Transformation

Executables (or programs) platform independent

OUTPUT WORKFLOW

EXECUTABLE WORKFLOW



Stage-in Job

Transfers the workflow input data

Cleanup Job

Removes unused data

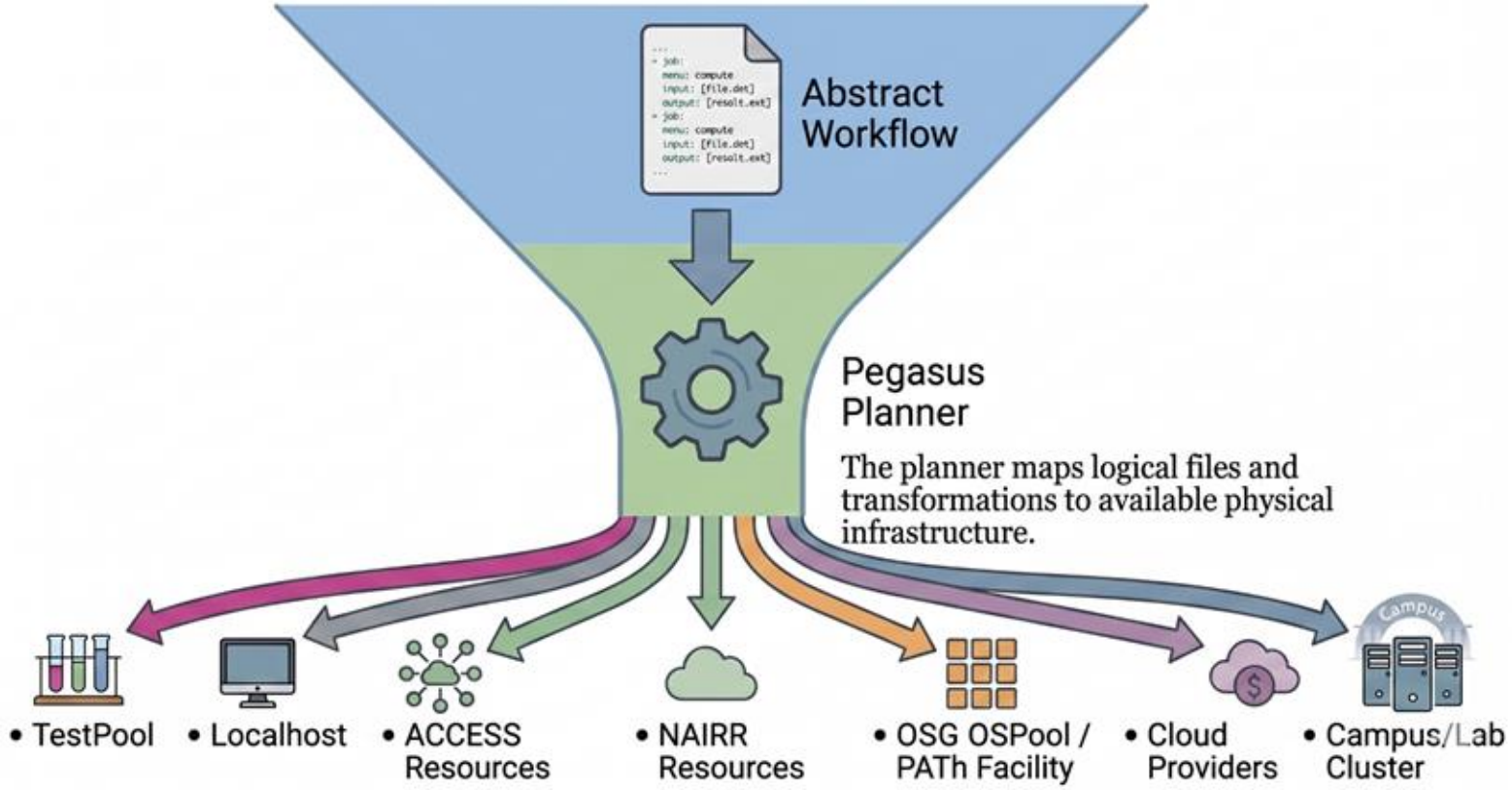
Stage-out Job

Stage-out generated output data

Registration Job

Registers the workflow output data

Write Once, Run Anywhere



Requirement: Any HPC/cloud/… system with outbound network connectivity.
Includes your own campus cluster or lab machines!

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Resilience: When Things Go Wrong



Automatic Retries

Jobs are automatically retried upon failure.

Checkpointing

Workflow-level checkpointing saves progress. Never restart from scratch.

Rescue DAGs

Generates a plan containing only the remaining work after a crash.

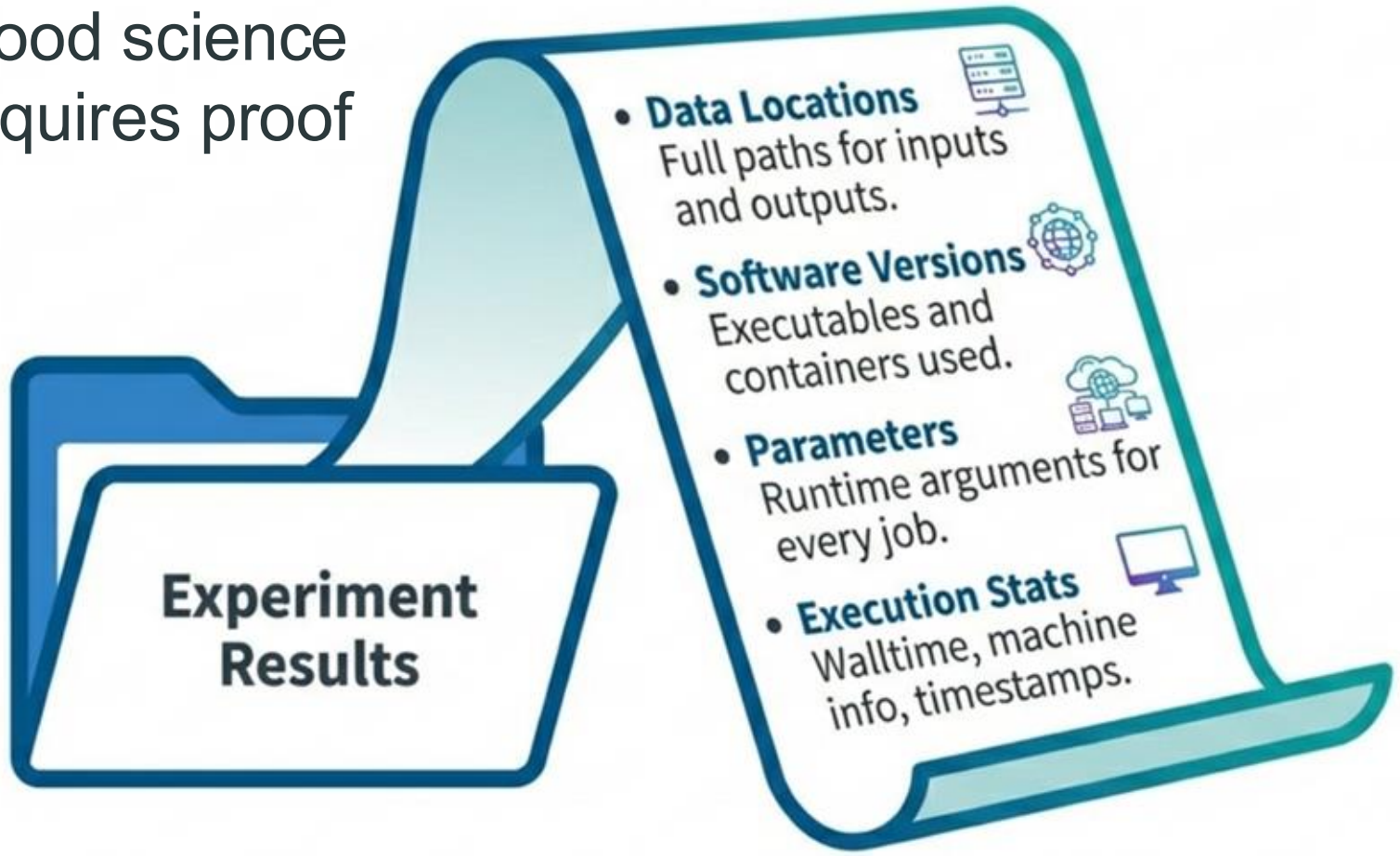
Debug Tools

`pegasus-analyzer` pinpoints specific errors (data vs. code).

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Provenance & Reproducibility

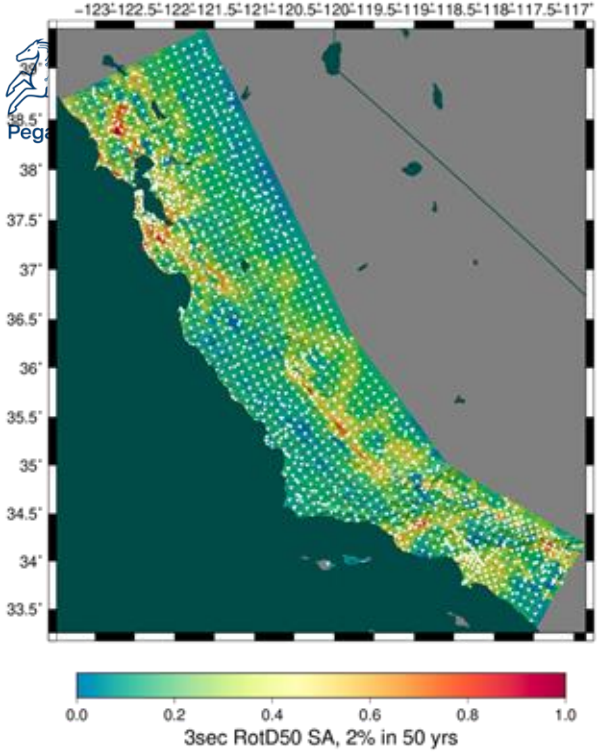
Good science requires proof



Pegasus tracks every bit of **data and every compute** step, ensuring your results are **verifiable** and **reproducible**.

Success Stories

Pegasus for Seismic Hazard Applications



45
Days
Duration of 24/7 operation

55K
Frontier
Node-hours
GPUs

125K
Frontera
Node-hours
CPUs

*High-water marks were 4130 nodes on Frontier (44%) and 1029 nodes on Frontera (12%)**

1 PB
Data
Managed by Pegasus

9M
Files
Staged to USC CARC

945
Workflows
27,720 jobs

Useful information for:

- ✓ Building engineers
- ✓ Disaster planners
- ✓ Insurance agencies

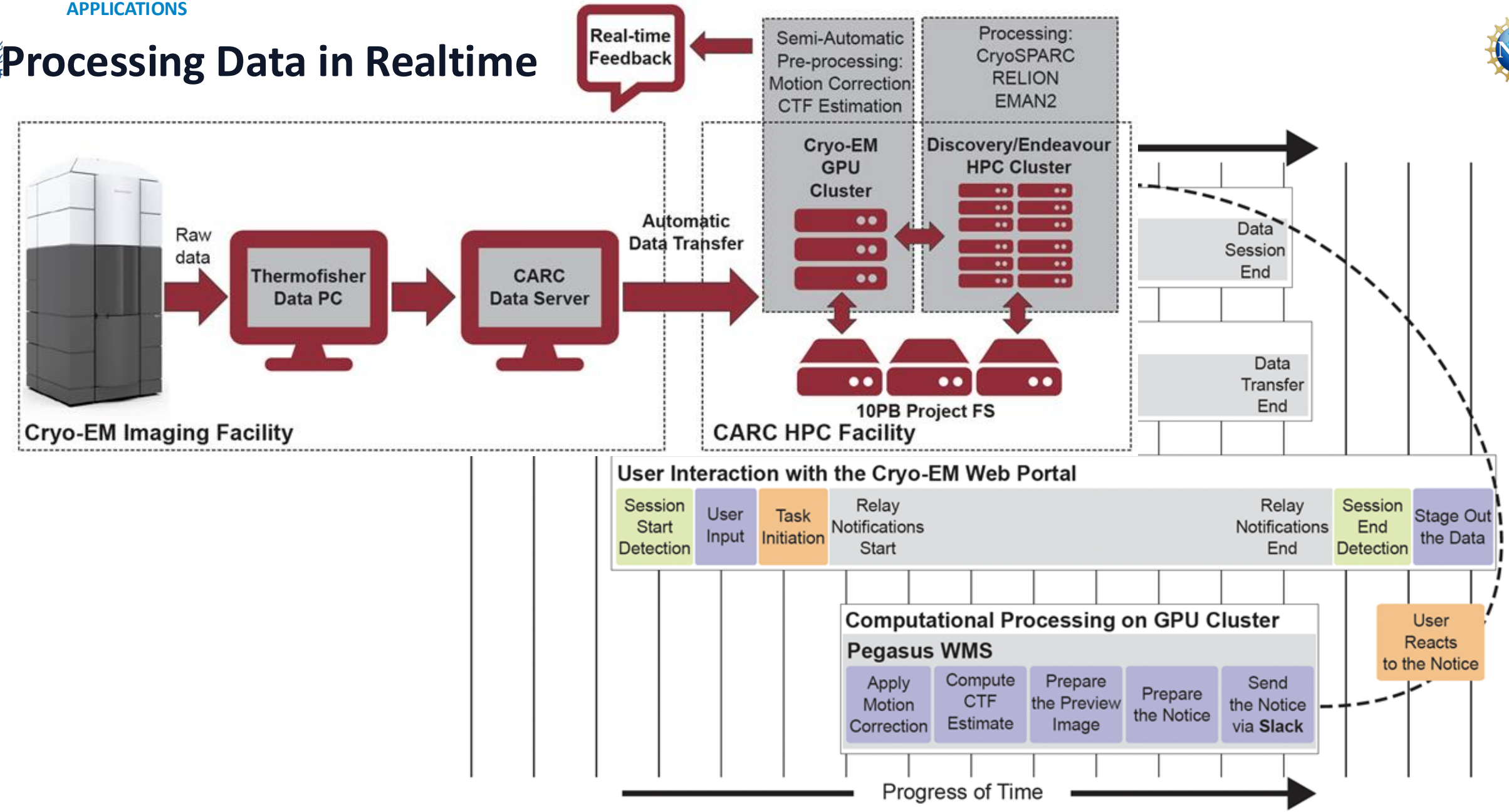
Two new Northern California hazard models were produced.

Slide Courtesy of Scott Callaghan, USC

* *Neither used a reservation or priority bump.*

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Processing Data in Realtime



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XENONnT - Dark Matter Search



Two Workflows

Monte Carlo simulations and the main processing pipeline.

- Workflows execute across Open Science Grid (OSG) & **European Grid Infrastructure (EGI)**
- Rucio for data management
- MongoDB instance to track science runs and data products.



```
------------------------------------------------------------------------------
Type           Succeeded Failed  Incomplete  Total     Retries   Total+Retries
Tasks          4000      0       0           4000      267       4267
Jobs           4484      0       0           4484      267       4751
Sub-Workflows  0         0       0           0         0         0
------------------------------------------------------------------------------

Workflow wall time                                       : 5 hrs, 2 mins
Cumulative job wall time                                 : 136 days, 9 hrs
Cumulative job wall time as seen from submit side        : 141 days, 16 hrs
Cumulative job badput wall time                          : 1 day, 2 hrs
Cumulative job badput wall time as seen from submit side : 4 days, 20 hrs
```


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APPLICATIONS

Event Horizon Telescope Bringing Black Holes into Focus

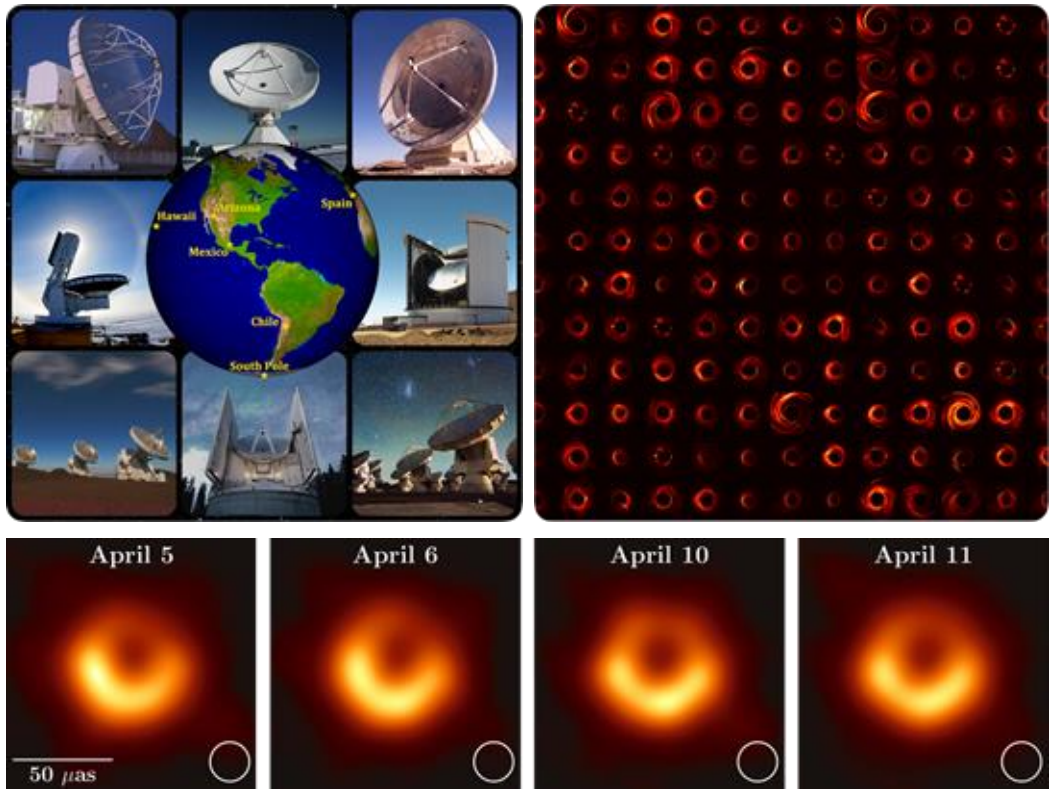
480,000 jobs - 2,600,000 core hours

#15 in all OSG projects in last 6 months

#2 in all OSG astronomy projects in the last 6 months

8 telescopes: 5 PB of data

60 simulations: 35 TB data

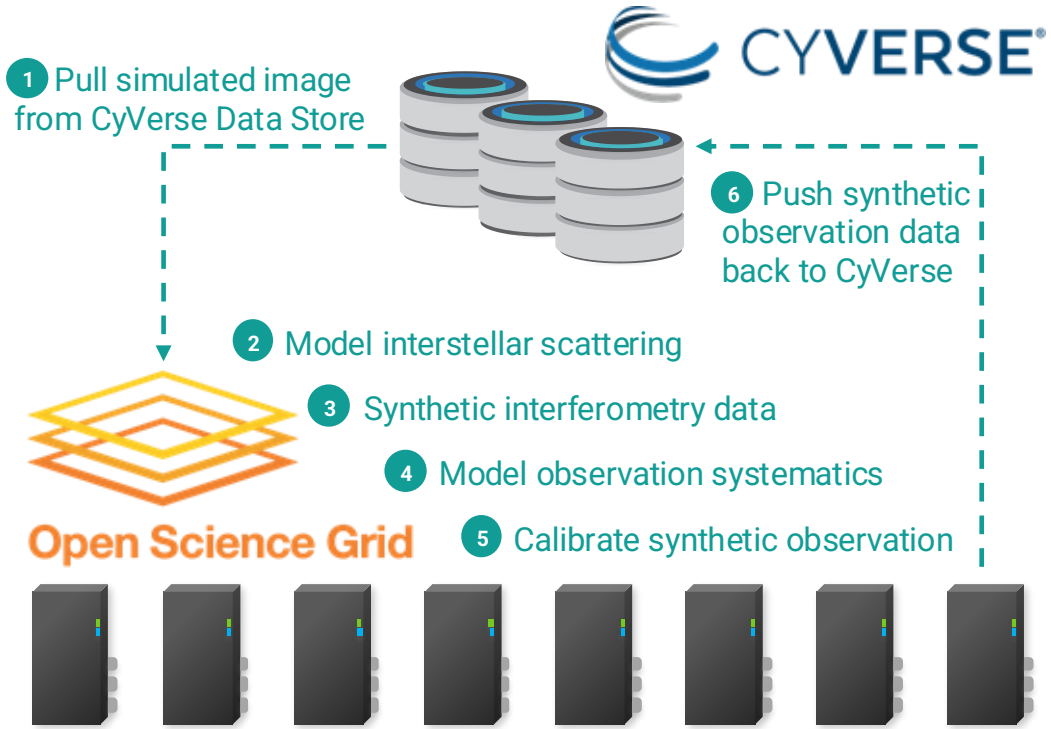


First images of black hole at the center of the M87 galaxy

Improve constraints on Einstein's theory of general relativity by 500x

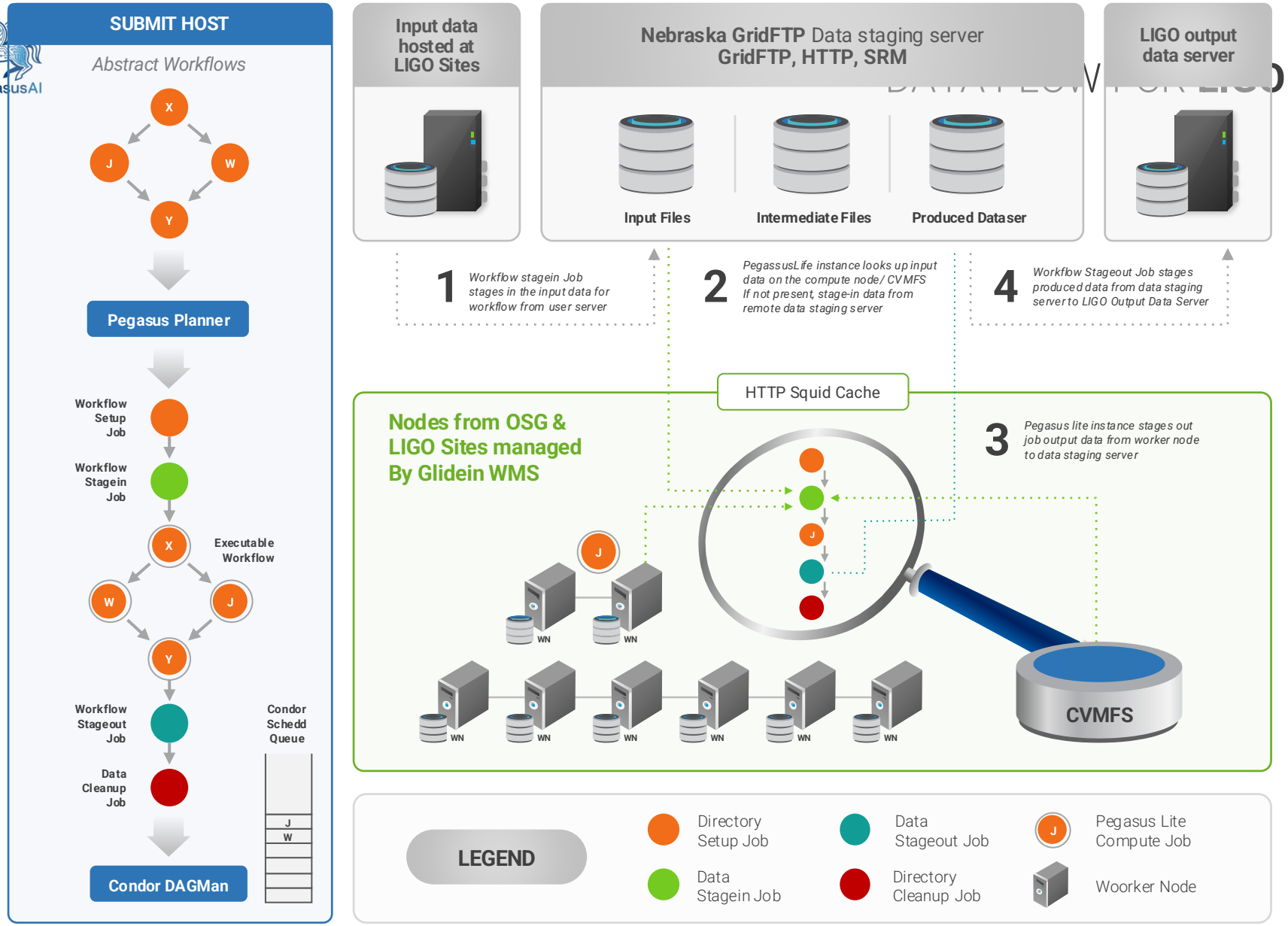
Pegasus-SYMBA Pipeline

Physically accurate synthetic observation data from simulations are keys to develop calibration and imaging algorithms, as well as comparing the observation with theory and interpreting the results.



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PEGASUS WORKFLOWS
Advanced LIGO IN OSG



60,000 Compute Tasks

Input Data: 5000 files (10GB total)

Output Data: 60,000 files (60GB total)

Processed Data: 725 GB

Laser Interferometer
Gravitational Wave Observatory

First Detection Runs executed on **LIGO Data Grid, EGI, Open Science Grid and XSEDE**

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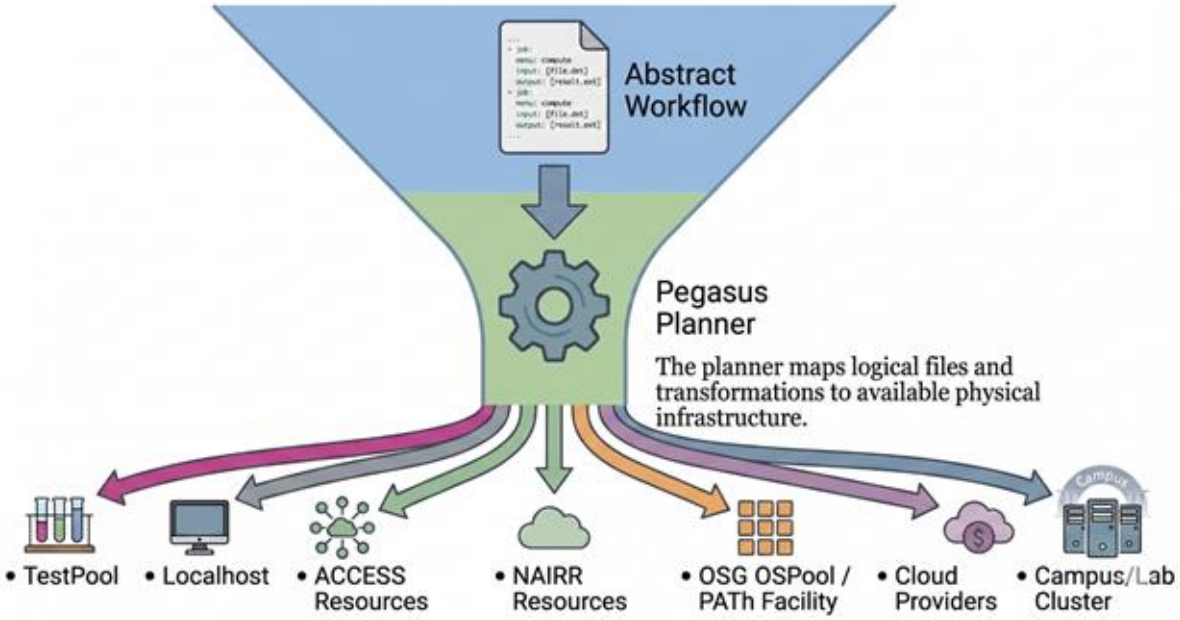
Try Pegasus?

An **easy and convenient** way to try out Pegasus and submit workflows to Laguna is to use the **Laguna Open OnDemand Instance.** https://laguna-ood.carc.usc.edu/

Hosted, web-based
Open OnDemand and Jupyter, with no local install.

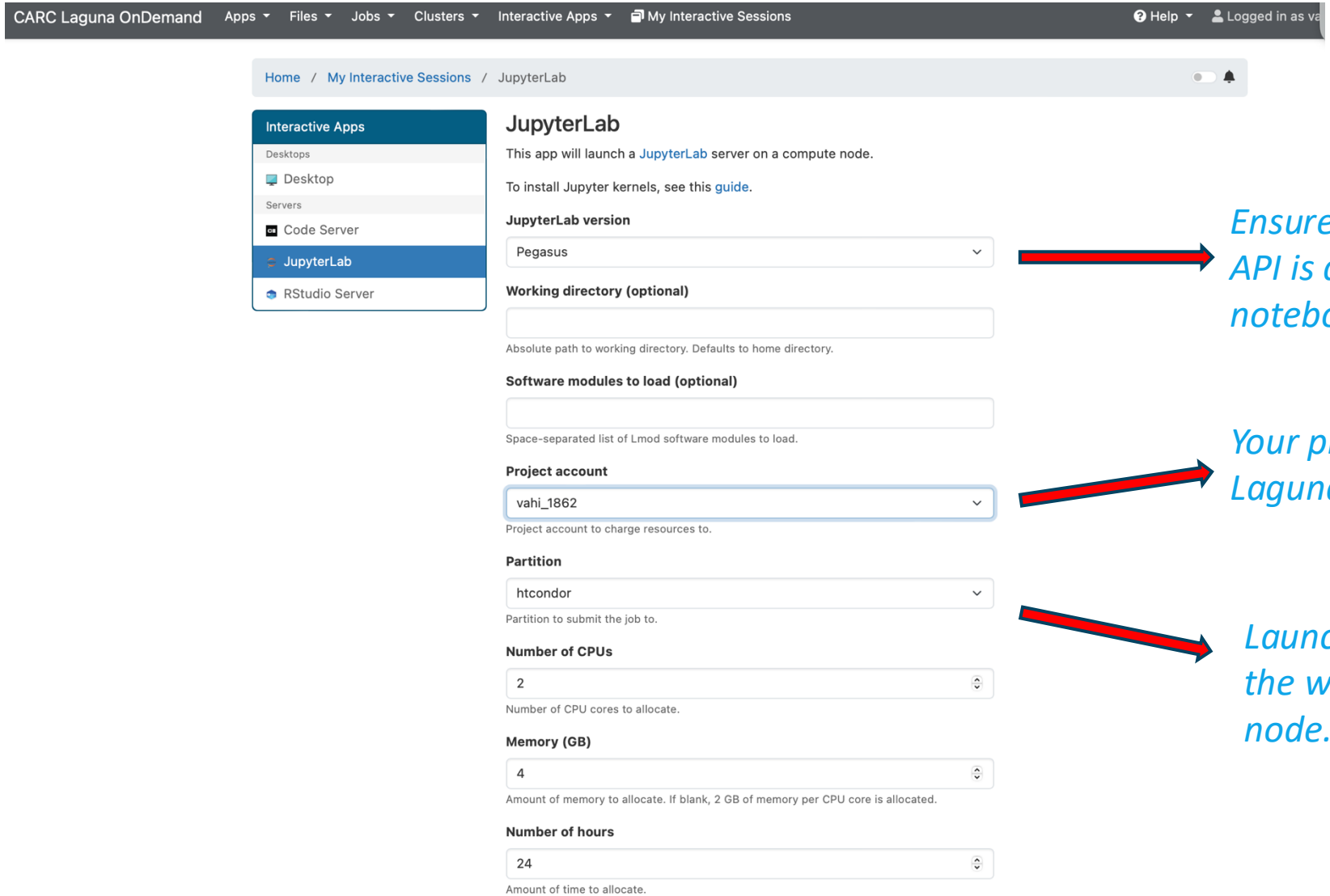
Single sign-on with your University Credentials

Training Notebooks
Try out Pegasus training examples
https://github.com/pegasus-isi/ACCESS-Pegasus-Examples



Log in: **pegasus.access-ci.org**
Help: support.access-ci.org/tools/pegasus

LAGUNA OPEN ONDEMAND – LAUNCH WORKFLOWS



Ensures Pegasus Workflow API is accessible via notebooks.

Your project account on Laguna

Launch the notebook on the workflow submission node.

We are here to help!

- Slack channel: find at https://pegasus.isi.edu/contact/
- Email: pegasus-support@isi.edu
- **Office Hours**

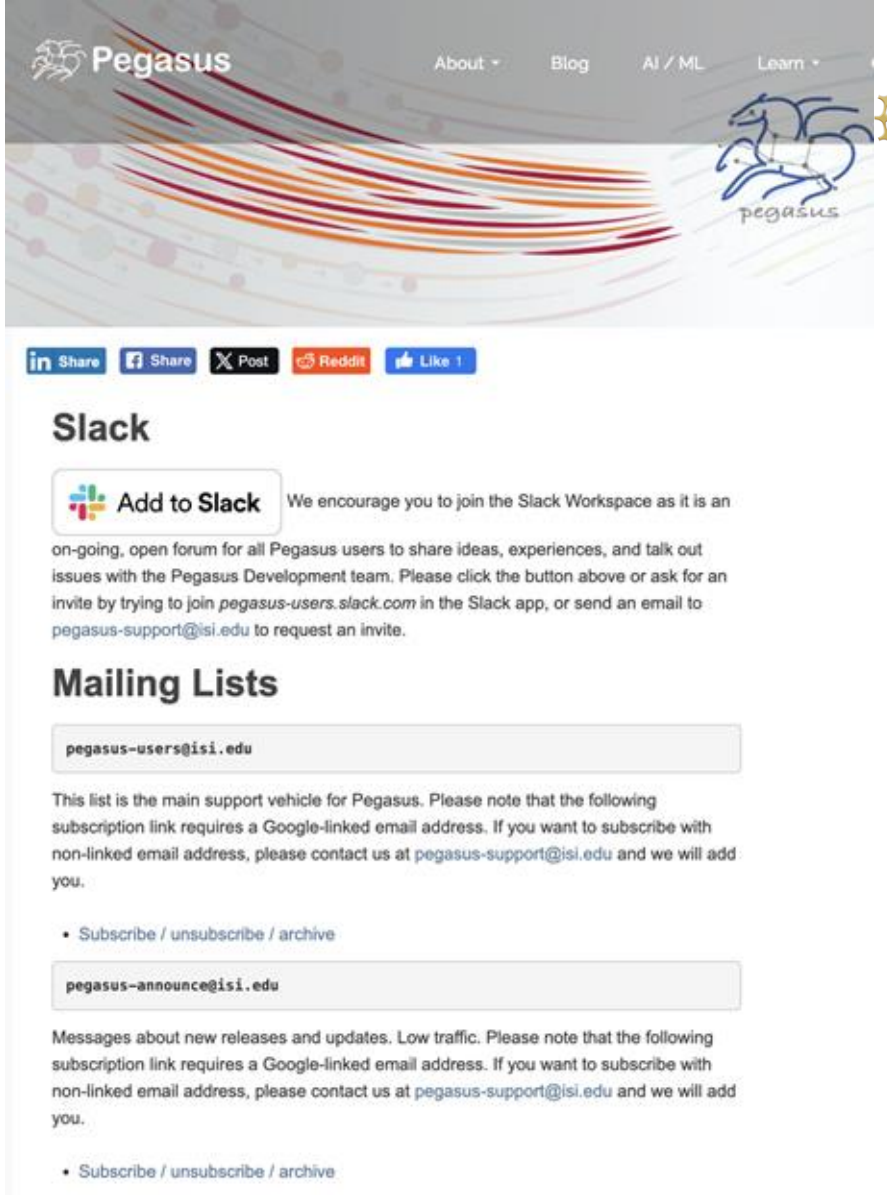
Join the Pegasus team every Friday for virtual office hours at 11 AM Pacific / 2 PM Eastern.

Do you have questions about workflows or need guidance on organizing and implementing them? Join our weekly office hours – designed to support both new and experienced users in learning and engaging with Pegasus. Here's what to expect:



ACCESS

- Tutorial walkthrough first Friday of the month
- https://pegasus.isi.edu



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Thank you!
Questions?

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Research Computing Facilitator. CARC. USC

BH **Bret Harry**
Systems Administrator. ITS. USC.

JH **James Hong**
Cold Front Developer. CARC. USC